

Geen

Assignment-04
Sheet-07 (DSA)

(22BC01068)

02/12/23

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- Sec-A ($2 \times 5 = 10$)

Q1. Define a complete binary tree.

Ans

A complete binary tree is a binary tree in which all level except last level are complete filled and all nodes in last level are as left as possible.

Q2. What is the difference among three types of traversals in a tree?

Ans

Inorder traversal: Visit the left subtree, then the root and finally the right subtree. This results in visiting nodes in ascending order in a binary search tree.

- Pre-order traversal: - Visits the root, then the left subtree and finally the right subtree. It can be used to create a copy of the tree.

- Post order traversal: Visit the left subtree, then the right subtree, and finally the root. It is often used in deleting nodes from a tree.

Q3.

What is the key rule of BST?

Ans

The key rule of a Binary Search tree is that for each node, all nodes in its left subtree have values less than the node's value, and all the nodes in its right subtree have values greater than the node's value.

Q4.

Why do we call AVL tree a self-balancing tree?

Ans

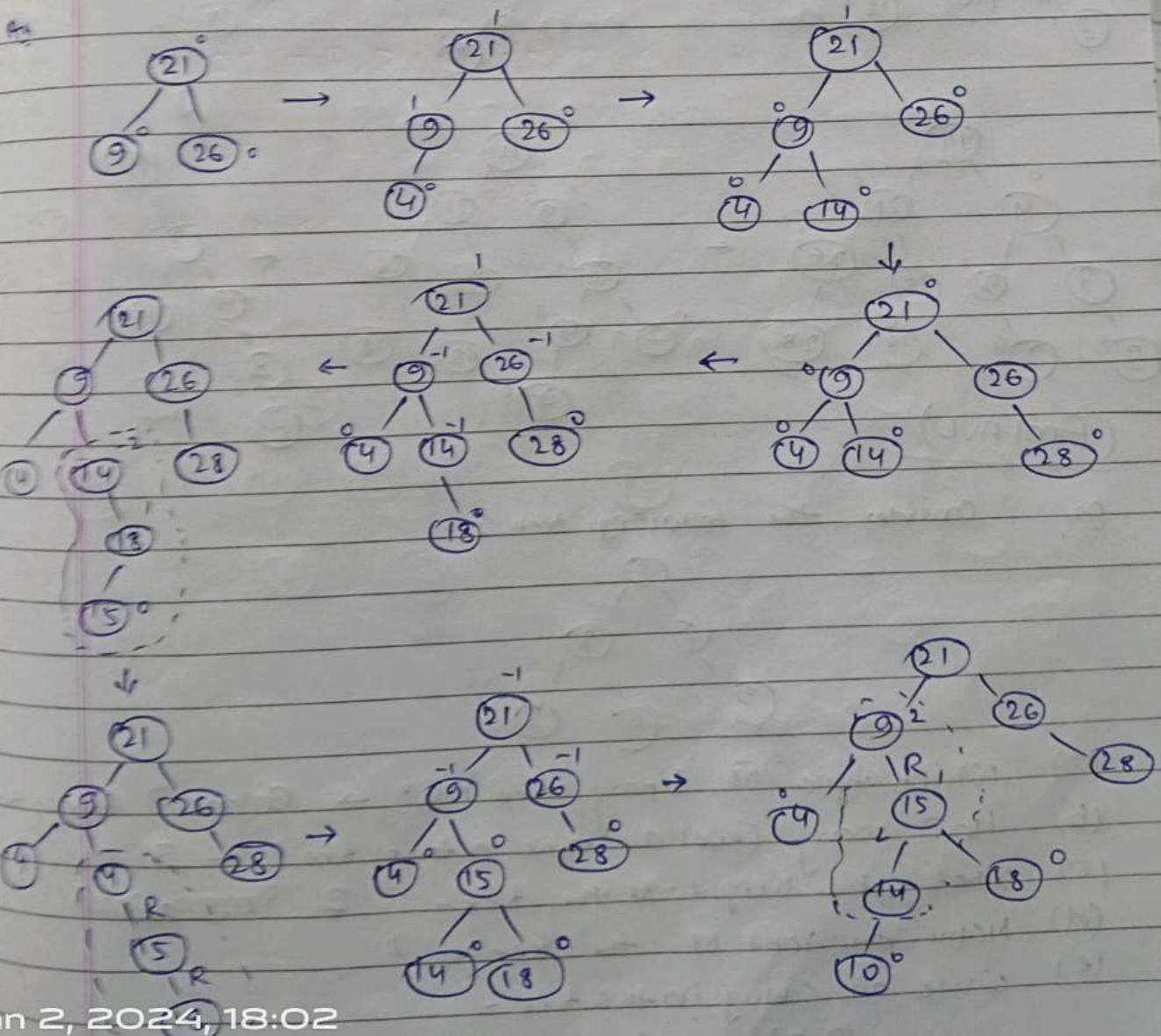
Because each node has a balance factor of 0, 1, -1 in AVL tree.

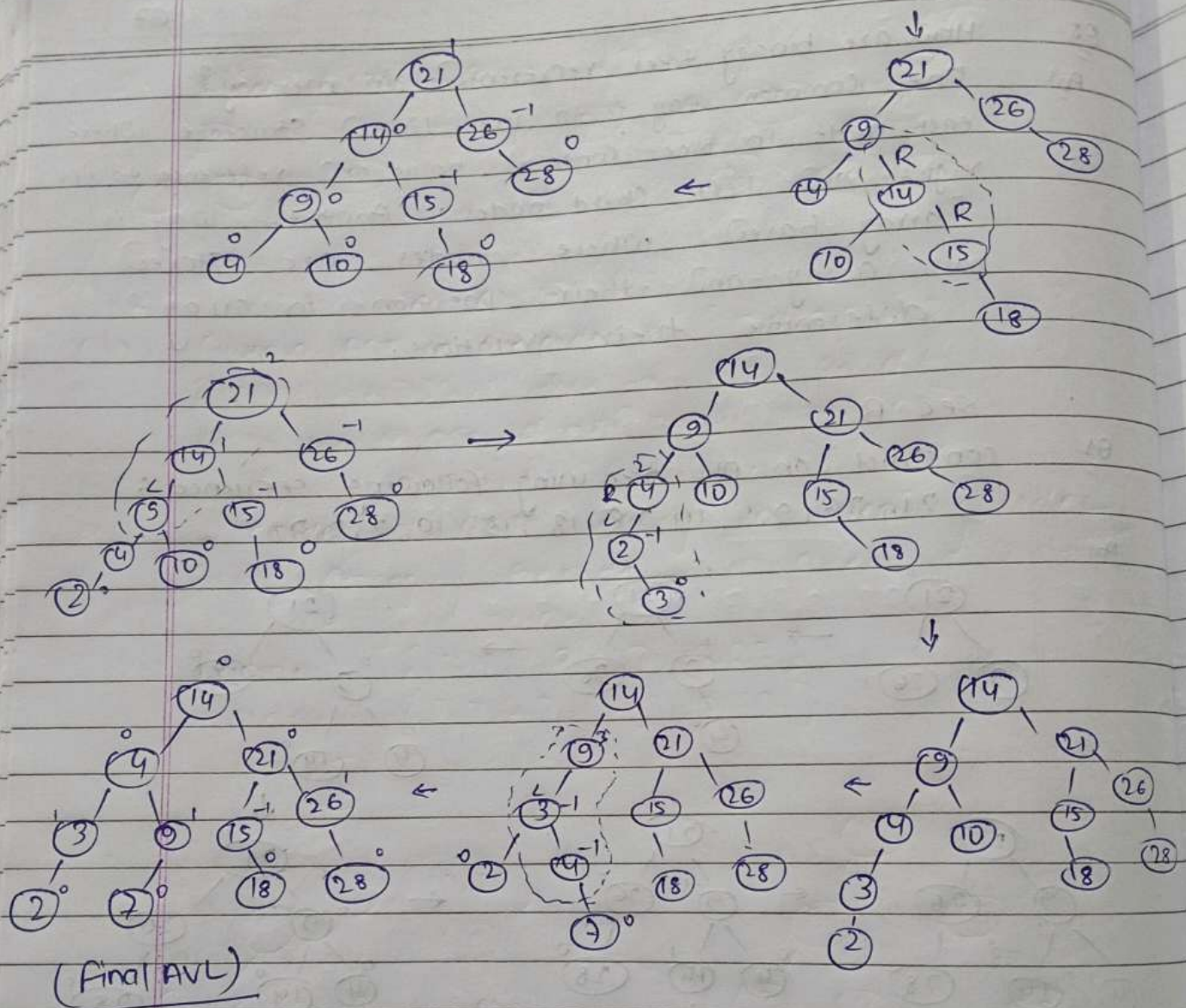
Q5. How are binary trees represented in memory?

Ans. One common way is to use linked structure where each node in tree contains value and reference to its right and left child nodes. Another way is array based, where nodes are stored in array and their position in array determine their relations.

Sec-B

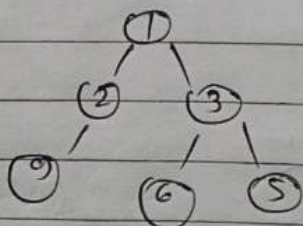
Q1. Construct an AVL tree using following sequence:
21, 26, 9, 4, 14, 28, 18, 15, 10, 2, 3, 7.





(Final AVL)

Q2. Consider the following tree:



- which type of binary tree it is? → Balanced Binary tree
- How many level are there in this tree? → 3 (0, 1, 2)
- what is the height of the tree - 3 (~~2, 1, and 2~~)
- Name the root of the tree - 1
- what are the sibling in this tree - (2, 3) and (9, 6, 5)

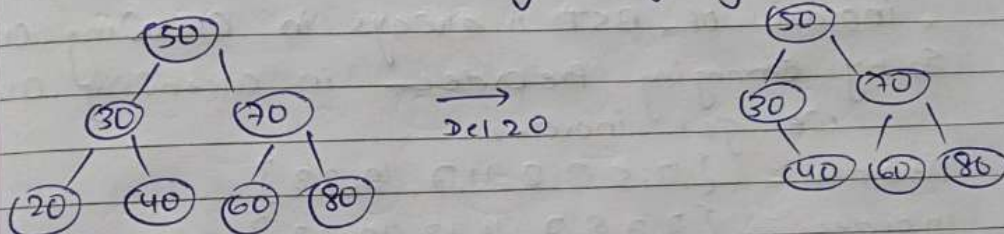
(f) Name all the internal and external nodes of this tree.
 internal nodes - 2, 3
 external nodes - 9, 6, 5

(g) Write the Post-order traversal of this tree.
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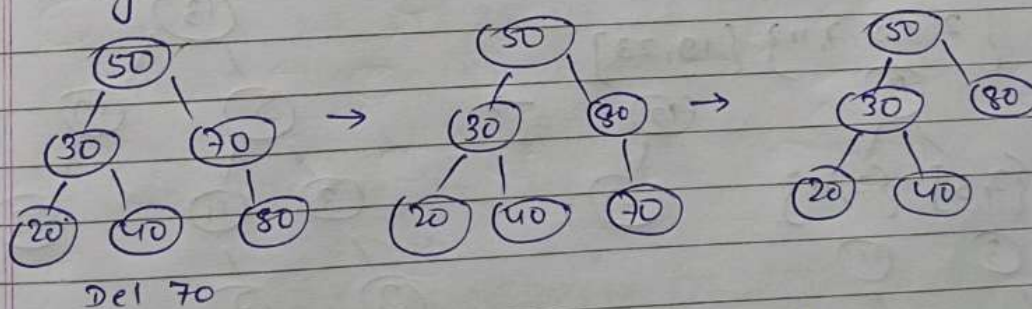
Discuss all the cases of BST deletion with example.

(1) Delete a Leaf Node :-

it can be done by assigning node to null

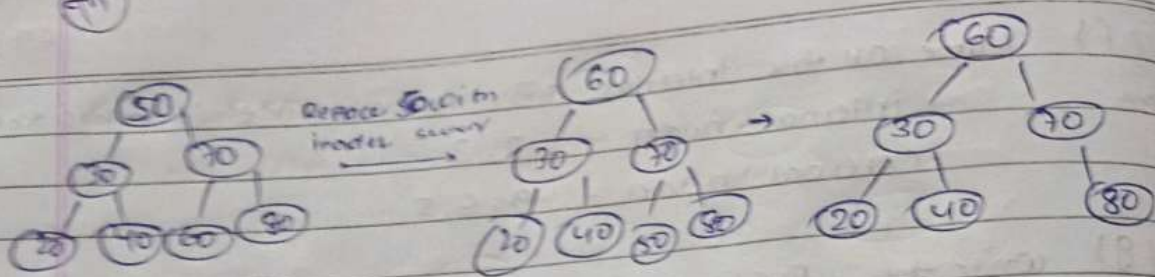


(2) Delete node with single child
 copy child to node and delete node



(3) Delete node with both side child
 The trick is to find inorder or inorder
 contents of node copy them to node
 and then simply delete

sequence



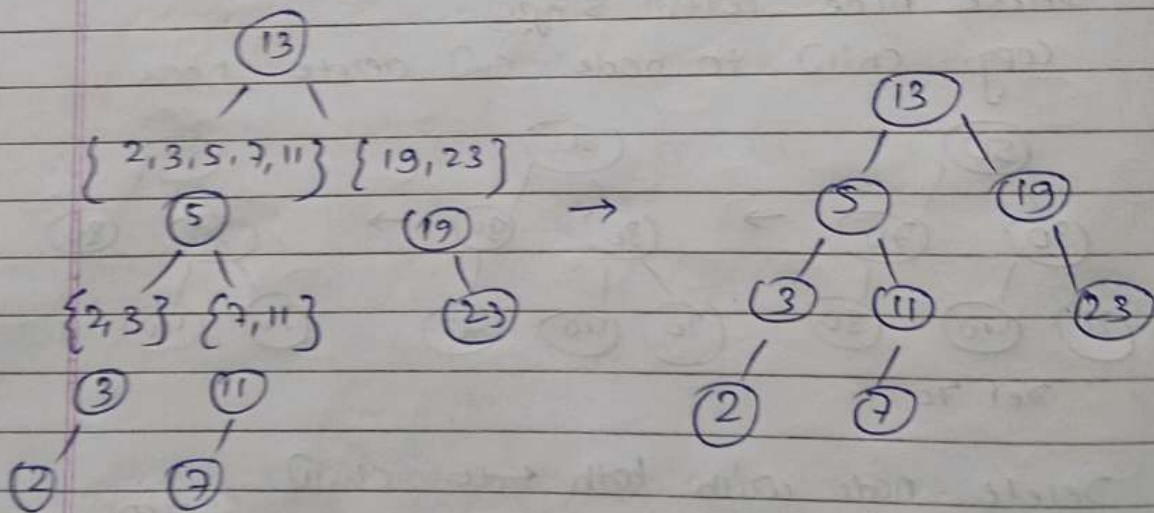
SEC-C

- (1) Given a BST in Pre order as $\{13, 5, 3, 2, 11, 7, 19, 23\}$
 $\{13, 5, 3, 2, 11, 7, 19, 23\}$ draw this BST and determine if this
 BST same as one derived in Post order as
 $\{2, 3, 5, 7, 11, 13, 19, 23\}$ $\{2, 3, 5, 7, 11, 23, 19, 13\}$.

Ans Inorder of BST is always in ascending order
 So by arranging Preorder in ascending order
 we get inorder.

Preorder = $\{13, 5, 3, 2, 11, 7, 19, 23\}$

Inorder = $\{2, 3, 5, 7, 11, 13, 19, 23\}$



Postorder -

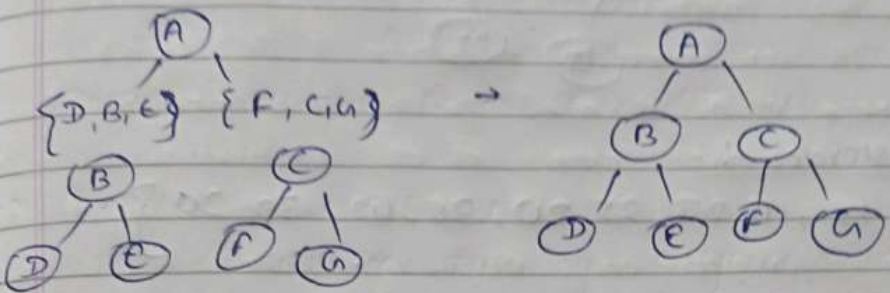
$\{2, 3, 7, 11, 5, 23, 19, 13\}$

Postorder for this BST is not same.

Q. Draw a binary tree using following traversal sequence:
in-order: D, B, E, A, F, C, G

post-order: D, E, B, F, G, C, A

Find its Preorder traversal sequence.



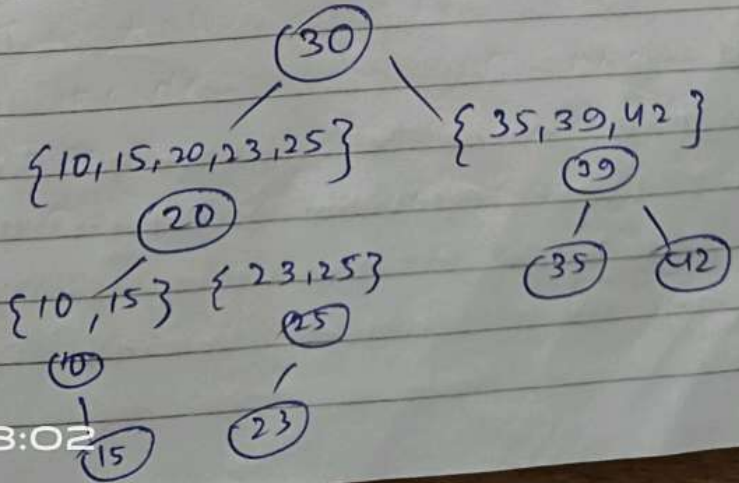
Preorder - ABDECFG

Q3. The Preorder traversal sequence of a binary search tree is -
30, 20, 10, 15, 25, 23, 39, 35, 42

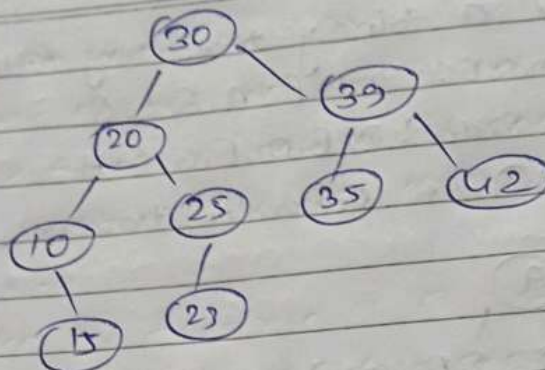
Which one of the following is the Post-order traversal sequence of the same tree?

- (1) 10, 20, 15, 23, 25, 35, 42, 39, 30
- (2) 15, 10, 25, 23, 20, 42, 35, 39, 30
- (3) 15, 20, 10, 23, 25, 42, 35, 39, 30
- (4) 15, 10, 23, 25, 20, 35, 42, 39, 30

Ans. Pre order = { 30, 20, 10, 15, 25, 23, 39, 35, 42 }
in order = { 10, 15, 20, 23, 25, 30, 35, 39, 42 }



Final BST -

Postorder :- $\{ 15, 10, 23, 25, 20, 35, 42, 39, 30 \}$